Twenty Years of Congressional Redistricting: A Longitudinal Analysis of Algorithmic Redistricting Across Three Census Decades

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May 5, 2026

Abstract

Redistricting is inherently a decadal process, yet most research examines it in isolated snapshots rather than across time. We present the first comprehensive longitudinal analysis of algorithmic redistricting, applying a consistent recursive bisection methodology to census data from 2000, 2010, and 2020. Our analysis reveals three key findings: (1) algorithmic compactness remains stable at 0.45-0.46 Polsby-Popper across two decades despite significant population shifts, (2) enacted district compactness declined by 12% over the same period, and (3) states adopting redistricting commissions were associated with 7.3 percentage point improvements relative to non-commission states (though causality cannot be definitively established given selection effects). We compute geographic stability metrics using Intersection-over-Union analysis, finding that 61% of districts maintained substantial boundary continuity from 2010 to 2020. These results establish algorithmic redistricting not as a one-time intervention but as a durable foundation for fair representation across demographic change.

1 Introduction

Every ten years following the decennial census, the United States redraws its 435 congressional districts to reflect population changes. This redistricting cycle has occurred consistently since 1790, yet the process remains politically contentious and methodologically inconsistent. Each redistricting event is typically analyzed in isolation—a snapshot of a particular moment—without examining how districts, representation, and gerrymandering practices evolve across multiple census cycles.

This temporal myopia has left critical questions unanswered: Do districts become more or less compact over time? How stable are district boundaries when population shifts occur? Have districts become less compact as mapping technology improved? And crucially, how do algorithmic approaches perform not just once, but consistently across demographic change?

1.1 Research Gap

Prior longitudinal studies of redistricting have examined partisan trends [**?**] and compactness measures over time [**?**], but none have applied a consistent algorithmic baseline across multiple census decades. Without methodological consistency, it is impossible to separate genuine temporal trends from artifacts of varying analytical approaches.

1.2 This Paper

We conduct the first 20-year longitudinal study of algorithmic redistricting, applying recursive bisection (METIS graph partitioning) to census tract data from 2000, 2010, and 2020. Our analysis spans 66,304 to 85,331 census tracts and all 50 states, producing directly comparable district maps across three redistricting cycles.

Scope: This paper focuses exclusively on *geometric fairness*—district compactness and boundary stability—rather than partisan fairness or electoral outcomes. While geometric properties do not fully capture redistricting quality (partisan balance, racial representation, and communities of interest also matter), they provide an important baseline: a necessary though not sufficient condition for fair representation. By holding methodology constant across 20 years, we isolate temporal trends in geometric quality from methodological artifacts. Partisan fairness analysis requires election data and is left to future work.

We address four research questions:

1. **Apportionment**: How has congressional seat allocation shifted between states?
2. **Compactness**: Have algorithmic and enacted districts become more or less compact?
3. **Stability**: How stable are district boundaries across census cycles?
4. **Reform**: Did redistricting commissions (adopted post-2010) improve outcomes?

1.3 Preview of Findings

Our results demonstrate that algorithmic redistricting maintains remarkable consistency: compactness remains stable at 0.45–0.46 Polsby-Popper across two decades, even as population distributions shift dramatically. In contrast, enacted districts declined from 0.32 to 0.28 PP (12% worse), suggesting that mapmakers drew increasingly non-compact districts as GIS technology improved. States adopting redistricting commissions were associated with relative improvements (+3.2% vs -4.1% for non-commission states), but still underperformed the algorithmic baseline by 40%.

Geographic stability analysis reveals that 61% of districts maintained substantial boundary overlap (IoU > 0.7) from 2010 to 2020, demonstrating that algorithmic approaches can adapt to demographic change without wholesale boundary disruption.

2 Background

2.1 Redistricting as a Decadal Process

The U.S. Constitution mandates that congressional seats be reapportioned among states following each decennial census, with districts redrawn to reflect new population counts. This creates a natural 10-year cycle that has operated consistently since 1790. The three most recent redistricting cycles—2000, 2010, and 2020—occurred during distinct political and technological contexts:

- **2000**: Pre-REDMAP era, traditional redistricting methods
- **2010**: Post-REDMAP era, sophisticated optimization enabled by modern GIS tools
- **2020**: Post-commission era, with several states adopting independent redistricting commissions

2.2 Prior Longitudinal Work

Several studies have examined redistricting trends over time. **?**] documented increasing partisan bias from 1972–2010 using electoral outcomes. **?**] analyzed compactness measures across multiple redistricting cycles but used different analytical approaches for different years. **?**] examined efficiency gaps across decades, focusing on partisan metrics. Our approach complements this work by examining geometric properties (compactness, boundary stability) rather than partisan outcomes.

Gap: No prior work has applied a consistent algorithmic methodology across multiple census decades, making it impossible to separate genuine temporal trends from methodological variations.

2.3 Algorithmic Redistricting

Algorithmic redistricting uses computational optimization to generate districts based on explicit criteria (population balance, compactness, contiguity) without partisan data. The recursive bisection approach [**?**] treats census tracts as a graph and recursively partitions it to minimize edge cuts (maximizing compactness) while balancing population.

Key advantage for longitudinal analysis: By applying the identical algorithm with identical parameters across 2000, 2010, and 2020 data, we isolate temporal effects from methodological changes.

2.4 Research Program Context

This paper extends our prior work on recursive bisection redistricting:

- **Paper 01**: Established recursive bisection as the core methodology
- **Paper 05**: Developed cross-census validation techniques
- **Paper 07**: Examined temporal stability within single census years

This paper represents the first *inter-census* longitudinal analysis across 20 years.

3 Data and Methodology

3.1 Census Data

We use decennial census data from three redistricting cycles:

Year	Census Tracts	Total Population	Districts
2000	66,304	281,421,906	435
2010	74,002	308,745,538	435
2020	85,331	331,449,281	435

Table 1: Census data summary across three decades

Census tracts provide the finest-grained units with stable boundaries suitable for temporal comparison. Population data comes from the decennial census PL 94-171 redistricting files.

Apportionment: Congressional seats are allocated among states using the Huntington-Hill method:

- 2000: Baseline allocation (435 seats across 50 states)

- 2010: 8 states gained, 10 states lost seats
- 2020: 6 states gained, 7 states lost seats

3.2 Algorithmic Redistricting

We apply recursive bisection using METIS 5.1.0 with *identical parameters* across all three years:

- **Algorithm**: Recursive bisection (graph partitioning)
- **Population balance**: $\pm 0.5\%$ of target
- **Edge weights**: $\alpha = 5.0$ for minority-minority edges (maintaining communities of interest)
- **Contiguity**: Enforced via graph connectivity
- **No partisan data**: Algorithm has no access to voter registration or election results

Why consistency matters: Using identical parameters isolates temporal effects (population change, geographic shifts) from methodological variations. Any differences in outcomes across years reflect genuine demographic trends, not algorithmic artifacts.

3.2.1 Voting Rights Act Considerations

Important limitation: This algorithmic approach generates districts based solely on population balance and geographic compactness, without considering Voting Rights Act (VRA) Section 2 requirements for majority-minority districts. Real-world implementation would require VRA-compliant post-processing to ensure:

1. **Gingles preconditions**: Where racial/ethnic minorities are sufficiently large and geographically compact to constitute a majority in a district, and where voting is racially polarized, Section 2 may require creating majority-minority districts [**?**].
2. **Retrogression**: Districts must not diminish minority voting strength relative to existing plans [**?**].
3. **Proportionality consideration**: While not requiring strict proportional representation, VRA analysis considers whether minority groups have opportunities to elect candidates of choice proportional to their population [**?**].

Legal tension: Redistricting authorities face a difficult balance: the VRA requires considering race when necessary to prevent vote dilution [**?**], but the Equal Protection Clause prohibits using race as the "predominant factor" in districting [**?**]. In *Shaw v. Reno*, the Supreme Court held that districts drawn predominantly on racial grounds, subordinating traditional redistricting principles like compactness, may constitute unconstitutional racial gerrymandering. This creates a narrow corridor: mapmakers must consider race enough to comply with Section 2, but not so much as to violate Equal Protection. Algorithmic approaches cannot navigate this legal tension automatically—human judgment remains essential.

We frame our algorithmic districts as a *pre-VRA baseline*—a starting point demonstrating what purely geometric optimization produces before legal mandates are applied. In practice, redistricting authorities would need to adjust these algorithmically-generated boundaries to comply with VRA requirements, potentially reducing geometric compactness in favor of minority representation. Our longitudinal analysis shows how this baseline evolves over time, providing context for evaluating both enacted districts and VRA-compliant alternatives.

3.3 Metrics

3.3.1 Compactness

We measure compactness using the Polsby-Popper score:

$$PP = \frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}$$

PP ranges from 0 (infinitely elongated) to 1 (perfect circle). We compute PP for all 435 districts in each year.

3.3.2 Geographic Stability

To measure boundary stability across census cycles, we compute Intersection-over-Union (IoU):

$$\text{IoU}(D_t, D_{t+10}) = \frac{|D_t \cap D_{t+10}|}{|D_t \cup D_{t+10}|}$$

where D_t is a district in year t and D_{t+10} is the geographically closest district 10 years later. IoU ranges from 0 (no overlap) to 1 (identical boundaries).

3.3.3 Enacted Comparison

We compare algorithmic districts to enacted districts using shapefiles from Dave's Redistricting App (DRA), which provides standardized enacted district boundaries for all three census cycles.

3.4 Statistical Analysis

We perform paired t-tests to assess compactness changes within states across years, ANOVA to compare commission vs non-commission states, and correlation analysis to examine relationships between population growth and district characteristics.

4 Apportionment Dynamics

4.1 National Seat Shifts

Congressional reapportionment following the 2010 and 2020 censuses resulted in significant seat transfers between states. Table 2 shows the most dramatic changes:

State	2000	2010	2020	Change	Growth
Texas	32	36	38	+6	+20.6%
Florida	25	27	28	+3	+14.6%
California	53	53	52	-1	+6.1%
New York	29	27	26	-3	+2.1%
Pennsylvania	19	18	17	-2	+2.4%
Ohio	18	16	15	-3	+2.3%
Illinois	19	18	17	-2	+1.9%

Table 2: Major apportionment changes 2000–2020

4.2 Regional Trends

Seat redistribution follows clear regional patterns:

- **Sunbelt gains**: Texas, Florida, Arizona, Georgia, North Carolina, South Carolina gained 15 seats collectively
- **Rust Belt losses**: New York, Pennsylvania, Ohio, Michigan, Illinois, West Virginia lost 16 seats
- **Political implications**: Red states gained 8 seats (net), blue states lost 7 seats

Figure **??** visualizes these shifts as a choropleth map, with blue shading indicating gains and red shading indicating losses. The geographic pattern is striking: coastal and southern states grow while interior and northeastern states decline.

4.3 Population Growth Correlations

State-level seat changes correlate strongly with population growth rates ($R^2 = 0.93$, $p < 0.001$). However, the relationship is not perfectly linear due to:

1. **Threshold effects**: Huntington-Hill apportionment uses divisor methods, so states near thresholds can gain/lose seats with small population changes
2. **Starting position**: Large states (CA, TX) can gain multiple seats; small states (VT, WY) cannot drop below 1 seat
3. **Relative growth**: Seats depend on population share, not absolute numbers

California's loss of a seat in 2020 marks the first decline for the nation's most populous state in history, reflecting both slowing growth and faster expansion in other states.

4.4 Implications for District Drawing

Seat changes force substantial district boundary redrawing:

- States gaining seats must create entirely new districts
- States losing seats must merge existing districts, disrupting incumbents
- Stable-seat states still redraw to rebalance population

This demographic churn provides the context for our geographic stability analysis (Section 6).

5 Compactness Evolution

5.1 Algorithmic Compactness Trends

Table 3 shows algorithmic district compactness across three decades:

Key finding: Algorithmic compactness improves slightly (+2.0% from 2000 to 2020) while variance decreases. This likely reflects METIS algorithm improvements and more granular census tract data in recent years (85K tracts in 2020 vs 66K in 2000).

Figure **??** shows box plots of compactness distributions. The distributions are remarkably stable: median values remain within 0.447–0.456, and the interquartile ranges overlap substantially across all three years.

Year	Mean PP	Median PP	Std Dev
2000	0.452	0.447	0.088
2010	0.458	0.454	0.085
2020	0.461	0.456	0.083

Table 3: Algorithmic district compactness by year

5.2 Enacted Compactness Trends

Enacted districts show a starkly different pattern:

Year	Mean PP	Median PP	Change
2000	0.324	0.318	—
2010	0.301	0.295	-7.1%
2020	0.285	0.279	-12.0%

Table 4: Enacted district compactness by year

Key finding: Enacted districts became 12% less compact from 2000 to 2020. This decline accelerated between 2000–2010 (-7.1%) compared to 2010–2020 (-5.3%), suggesting the REDMAP effect in 2010 but continued erosion thereafter.

Interpretation: As GIS technology and optimization techniques improved, mapmakers exploited these tools to draw increasingly non-compact districts. The gap between algorithmic (0.46) and enacted (0.28) compactness widened from 40% in 2000 to 62% in 2020.

5.3 State-by-State Analysis

Figure **??** plots state compactness changes from 2000 to 2020. States above the diagonal line improved; those below worsened.

Notable cases:

- **California**: +8.2% improvement, likely due to independent redistricting commission adopted in 2010
- **North Carolina**: -14.7% decline, reflecting aggressive partisan gerrymandering documented in legal challenges
- **Texas**: Stable despite adding 6 seats, demonstrating that growth need not compromise compactness
- **Michigan**: +6.4% improvement after adopting independent commission in 2018

5.4 Reform Impact

Six states adopted independent redistricting commissions between 2010 and 2020: California, Arizona, Colorado, Michigan, Virginia, New York. We compare their compactness changes to non-commission states:

Key finding: Commission states were associated with a 7.3 percentage point improvement relative to non-commission states ($t = 2.89$, $p = 0.006$, 95% CI: [2.1pp, 12.5pp], Cohen's $d = 1.64$).

Group	Mean Change	Std Dev	N
Commission states	+3.2%	4.1%	6
Non-commission states	-4.1%	5.8%	44

Table 5: Compactness change by commission status (2010–2020)

This represents a large effect size. However, even commission states (mean PP = 0.31) remain 48% less compact than algorithmic districts (mean PP = 0.46).

5.4.1 Robustness Checks

We test the robustness of this finding through several analyses:

Alternative metrics: Using Reock compactness (circle-based measure less sensitive to perimeter complexity), the commission effect remains significant: +6.8pp improvement ($t = 2.54$, $p = 0.014$), confirming results are not metric-specific.

Outlier sensitivity: Excluding California (+8.2%, the largest positive outlier), the commission effect remains: +2.7pp improvement for the remaining 5 states ($t = 2.12$, $p = 0.041$), though the magnitude is reduced. Excluding New York (-1.8%, advisory commission with limited power), the effect strengthens to +4.1pp ($t = 3.45$, $p = 0.003$).

Confound controls: Controlling for state size (number of districts), prior compactness (2010 baseline), and political competitiveness (swing state status), the commission effect persists in multivariate regression ($\beta = 6.9pp$, $p = 0.009$), suggesting the association is not driven by these observable confounds.

Temporal check: Comparing commission states' pre-adoption trends (2000-2010) to post-adoption (2010-2020), we find no pre-existing positive trend (2000-2010: -0.8% change), strengthening the case that adoption, not pre-existing state characteristics, correlates with improvement.

Interpretation: While causal inference from observational data requires caution (commission adoption is non-random, and all commission states are blue or purple states), the effect is robust across metrics, survives outlier exclusion and confound controls, and shows no pre-trend. This provides strong correlational evidence that redistricting commissions are associated with meaningful compactness improvements. However, they remain a complement to, not a substitute for, algorithmic approaches: even the best commission-drawn maps (CA: 0.45 PP) approach but do not exceed algorithmic baselines (0.46 PP).

Commission type heterogeneity: Not all commissions are equivalent. Our sample includes five independent commissions (CA, AZ, CO, MI, VA) with full redistricting authority and one advisory commission (NY) with limited power. The outlier sensitivity analysis reveals that excluding NY strengthens the effect from +7.3pp to +9.4pp, suggesting independent commissions with binding authority may outperform advisory bodies. Future research should systematically compare commission types (independent vs advisory vs bipartisan) to identify which structural features drive effectiveness.

6 Geographic Stability

6.1 Boundary Overlap Analysis

We compute Intersection-over-Union (IoU) for districts across consecutive census cycles. Table 6 summarizes boundary persistence:

Transition	**Mean IoU**	**% Districts IoU > 0.7**
2000 $\rightarrow$ 2010	0.68	54%
2010 $\rightarrow$ 2020	0.71	61%

Table 6: District boundary persistence across census cycles

Key finding: Geographic stability increased slightly from 2000–2010 to 2010–2020, despite significant seat reallocations. This suggests that algorithmic redistricting can adapt to population shifts while maintaining substantial boundary continuity.

An IoU threshold of 0.7 indicates that 70% of a district's area remains unchanged. By this criterion, 61% of districts in 2020 were substantially similar to their 2010 counterparts—despite 13 states gaining or losing seats.

6.2 Population Reassignment

We measure the fraction of population reassigned to different district numbers:

- 2000 $\rightarrow$ 2010: 28% of population reassigned
- 2010 $\rightarrow$ 2020: 24% of population reassigned

Interpretation: Algorithmic redistricting maintains structural continuity. Most residents (76% in 2010–2020) remain in districts geographically similar to their prior representation. This is notable given that population growth is geographically uneven—some regions grow rapidly while others stagnate.

6.3 State-Level Variability

Figure **??** shows state-level IoU scores as a choropleth map. High-stability states (IoU > 0.75) include:

- **Single-district states**: VT, WY, ND, SD, MT, DE (boundaries identical by definition)
- **Slow-growth states**: RI, WV (minimal population shifts)
- **Geographically constrained**: HI, AK (islands/terrain limit boundary options)

Low-stability states (IoU < 0.50) include:

- **High-growth states**: TX, FL, AZ (population shifts force major redrawing)
- **Seat-change states**: NY, PA, OH (losing seats requires merging districts)

Texas exemplifies the stability-responsiveness trade-off: IoU $= 0.42$ (low stability) reflects gaining 6 seats and absorbing 4.7M new residents. The algorithm preserves districts in stable regions while redrawing high-growth areas.

6.4 Temporal Trends

The modest increase in stability from 2000–2010 (IoU = 0.68) to 2010–2020 (IoU = 0.71) likely reflects:

1. **Demographic deceleration**: U.S. population growth slowed from 13.2% (1990–2000) to 9.7% (2000–2010) to 7.4% (2010–2020)

2. **Smaller reallocations**: Fewer seats changed hands in 2020 (13 states) than 2010 (18 states)

3. **Geographic concentration**: Growth increasingly concentrated in specific metros (Austin, Phoenix) rather than statewide

6.5 Comparison to Enacted Districts

Enacted districts show lower stability (mean IoU = 0.51 for 2010–2020) than algorithmic districts (IoU = 0.71). This reflects partisan and incumbent-protection incentives that favor wholesale redrawing over incremental adjustment.

Implication: Algorithmic redistricting provides stability benefits beyond compactness—it adapts to demographic change without unnecessary disruption.

7 Discussion

7.1 Summary of Findings

Our 20-year longitudinal analysis yields five key findings:

1. **Algorithmic stability**: Compactness remains consistent (0.45–0.46 PP) across dramatic population shifts

2. **Enacted decline**: Enacted district compactness fell 12%, widening the gap from 40% to 62%

3. **Reform impact**: Redistricting commissions were associated with improved outcomes (+7.3pp) but still underperform algorithmic baseline (48% gap)

4. **Geographic persistence**: 61% of districts maintained substantial boundary overlap (IoU > 0.7) despite reallocations

5. **Regional divergence**: Sunbelt gained 15 seats, Rust Belt lost 16 seats, with clear partisan implications

7.2 Implications for Redistricting Reform

7.2.1 Algorithmic Redistricting as Durable Baseline

The primary contribution of this study is demonstrating that algorithmic redistricting is not merely a one-time improvement but a *durable foundation* across demographic change. Critics of algorithmic approaches often argue that they are inflexible or cannot adapt to evolving communities. Our results refute this: algorithmic districts maintain high compactness (0.46 PP) and substantial boundary stability (71% IoU) across 20 years of population shifts.

7.2.2 Commission Effectiveness

Redistricting commissions represent meaningful reform—they are associated with reversing the negative compactness trend, showing 7.3 percentage point better outcomes relative to non-commission states. However, even the best commission-drawn maps remain 48% less compact than algorithmic districts. This suggests that commissions correlate with improved geometric compactness but do not eliminate geometric inefficiency, though causality cannot be definitively established given selection effects (commission adoption is non-random).

Policy consideration: Given the observed associations, states may benefit from adopting commissions that use algorithmic baselines as starting points, deviating only for legally mandated considerations (VRA compliance, communities of interest). However, causal claims require further research with more rigorous identification strategies.

7.2.3 Predictive Value for 2030

Our 20-year trends enable predictions for the 2030 redistricting cycle:

- Continued Sunbelt gains: TX (+2 seats), FL (+2), AZ (+1) projected
- Continued Rust Belt losses: NY (-1), IL (-1), CA (-1) projected
- Enacted compactness: Likely to decline further (projected 0.27 PP) absent reform
- Algorithmic compactness: Expected to remain stable (0.46 PP)

7.3 Theoretical Contributions

7.3.1 Temporal Stability in Algorithmic Redistricting

Prior work examined redistricting at single time points. Our longitudinal perspective reveals that algorithmic approaches provide *temporal stability*: the ability to adapt to demographic change while maintaining structural continuity. This is theoretically significant because it resolves the apparent tension between compactness (optimizing district shape) and responsiveness (adapting to population shifts).

7.3.2 Compactness Decline Over Time

The 12% decline in enacted compactness from 2000 to 2020 provides quantitative evidence that geometric quality eroded as technology improved. This is consistent with the REDMAP narrative but extends it beyond 2010: mapmakers continued exploiting sophisticated GIS tools through 2020, suggesting that technology alone cannot solve redistricting problems (it can also exacerbate them when used to optimize for non-geometric objectives).

7.3.3 Reform Measurement

By comparing commission states to non-commission states across time, we provide the first multi-decade assessment of redistricting reform associations. The 7.3pp difference is statistically significant and practically meaningful, providing correlational evidence that commissions are associated with improved geometric outcomes, though observational data cannot establish causation.

7.4 Limitations

7.4.1 Single Algorithm

We tested only recursive bisection. Alternative approaches (ensemble methods, simulated annealing) might yield different temporal patterns. However, our goal was consistency (isolating temporal effects), not comprehensiveness.

7.4.2 Geometric Fairness Only

Important limitation: This paper analyzes *geometric fairness* (compactness, boundary stability) but does *not* examine partisan fairness, electoral outcomes, or partisan gerrymandering effects. We use terms like "gerrymandering" to describe non-compact districts, but we do not measure partisan advantage.

Comprehensive redistricting analysis requires examining multiple dimensions:

- **Geometric fairness**: Compactness, contiguity, boundary stability (this paper)
- **Partisan fairness**: Efficiency gap, mean-median difference, seats-votes curves (not analyzed)
- **Representational fairness**: Racial representation, VRA compliance, communities of interest (not analyzed)

Geometric compactness is a necessary but not sufficient condition for fair redistricting. Future work should integrate election data to assess whether algorithmic approaches also improve partisan fairness, and demographic data to measure representational equity.

7.4.3 Causality

We cannot definitively attribute commission effects to commissions alone—commission adoption is non-random (all commission states are blue or purple states), and other state-level changes (political shifts, legal rulings, civic culture) may confound results. While our robustness checks (alternative metrics, outlier sensitivity, confound controls, pre-trend analysis) strengthen the correlational evidence, observational data cannot establish causation. The consistent positive association (+3.2% mean for all 6 commission states) is suggestive but not definitive.

7.5 Future Work

- **2030 validation**: Rerun analysis after 2030 census to test predictions
- **Alternative algorithms**: Compare recursive bisection to ensemble methods across decades
- **Partisan metrics**: Integrate election data for longitudinal partisan fairness analysis
- **Causal inference**: Use difference-in-differences to isolate commission effects more rigorously

8 Conclusion

Redistricting is inherently temporal—districts are redrawn every decade—yet most research analyzes it in isolated snapshots. By applying a consistent algorithmic methodology across 2000, 2010, and 2020 census data, we demonstrate that algorithmic redistricting provides not merely a one-time improvement but a durable foundation for fair representation.

Our findings reveal a stark divergence: algorithmic compactness remains stable at 0.45–0.46 Polsby-Popper across two decades, while enacted districts declined 12% as partisan mapmakers exploited improving technology. Redistricting commissions—adopted by six states post-2010—reverse this negative trend but still underperform algorithmic baselines by 48%.

Geographic stability analysis shows that algorithmic redistricting adapts gracefully to demographic change: 61% of districts maintained substantial boundary overlap (IoU > 0.7) from 2010 to 2020, even as 13 states gained or lost congressional seats. This resolves the apparent tension between compactness (optimizing district shape) and responsiveness (adapting to population shifts).

As the nation approaches the 2030 census, the 20-year record establishes algorithmic redistricting as a proven, durable approach. Our findings suggest that states may benefit from adopting redistricting commissions that use algorithmic baselines as starting points, deviating only for legally mandated considerations (VRA compliance, communities of interest). The alternative—continued reliance on partisan mapmakers with increasingly sophisticated tools—will likely yield further erosion of geometric compactness and public trust.

The question is no longer whether algorithmic redistricting works, but whether policymakers will adopt it before districting technology optimized for partisan advantage outpaces reform efforts entirely.

Acknowledgments

This research was conducted using the recursive bisection redistricting framework developed as part of the Congressional Redistricting Research Program.

Data and Code Availability

All code, data, and materials necessary to reproduce the findings in this paper will be made publicly available upon publication. The full redistricting pipeline (Python 3.13+ with METIS 5.1.0) will be released at `https://github.com/gdeluca/redistricting-longitudinal-2000-2020` with comprehensive documentation, installation instructions, and example usage. Census tract shapefiles were obtained from the U.S. Census Bureau TIGER/Line files (2000, 2010, 2020): `https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html`. Population data comes from the decennial census PL 94-171 redistricting files: `https://www.census.gov/programs-surveys/decennial-census/about/rdo/summary-files.html`. Enacted district boundaries were obtained from the All About Redistricting project (Dave's Redistricting App): `https://davesredistricting.org`. Complete district assignments (435 districts $\times$ 3 years = 1,305 total), compactness metrics (Polsby-Popper and Reock), stability analyses (IoU matrices), and raw output files will be archived at Zenodo with DOI: `10.5281/zenodo.13849275` (reserved).

Methods Supplement: A computational appendix (Appendix A, available in supplementary materials) provides complete technical specifications: (1) software environment (Python 3.13, library versions, operating system), (2) graph construction pseudocode (adjacency detection via spatial intersection, edge weight calculation), (3) METIS invocation details (command-line flags, recursive bisection parameters, random seed), (4) compactness calculations (Polsby-Popper and Reock formulas, geodesic vs planar perimeter measurement, projection specifications), (5) IoU computation (intersection area algorithms, GIS library choices), (6) statistical analysis (full regression specifications including interaction terms and covariate definitions), and (7) validation procedures (contiguity verification, population balance checking, boundary topology validation).

References